An AI repair should come with evidence.

A candidate can print “success” and leave the bug behind. Giving generated code host access also gives a small task a large failure surface.

Reproduce the failure. Contain the repair. Check it independently. Keep evidence for a fresh replay.

The actual workflow

A proposal is not a passing result.

Reproduce The original implementation must fail independent cases.

Propose Vultr Serverless Inference produces a bounded Python repair.

Evaluate A fresh gVisor sandbox runs it; expected answers stay in the controller.

Remove & replay Confirm cleanup and export the candidate, checks and fingerprints.

Twenty independent cases for each maintained fixture: calendar rollover, integer-cent rounding and webhook retries.

Reference examples are disclosed model-free executions. Live repairs use actual Vultr inference.

Architecture and verified deployment

Vultr controls the complete run.

HTTPS browser > Vultr VM controller > Vultr Serverless Inference > disposable gVisor sandbox > independent evaluator > evidence.

Pinned runtime and image; no network, read-only root and mounts, nonroot identity.

Fixed CPU, memory, PID, time and streamed-output limits; bounded job admission and retry.

Genuine live repair: all 20 cases passed, fresh replay and cleanup verified.

The VM’s existing OptimAI crawler was preserved.

Deployed source: b5cbdb9ad95816d26bf56183d0e9972acbd5e469. Show actual containment evidence in the video.

Useful first scope and limits

Small failures. Inspectable evidence.

ReproLab makes agent-generated repairs reviewable and repeatable for three owned Python fixtures. It checks outputs and lifecycle, rather than trusting a success banner.

Arbitrary repositories and additional languages are outside this version.

The trusted controller and Docker administrator remain part of the boundary.

Integrity hashes bind recorded bytes; they are not hardware or host attestations.

Successful tests are evidence for these cases, not a security audit or universal correctness proof.